

Sprint 1 Document

UPRN Research

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1. Introduction

1. What the project is about

This project aims to develop a system capable of extracting street layouts and residential building data from OpenStreetMap (OSM) and using that information to design a proposed telecommunications network infrastructure. The overall objective is to create an algorithm that can automatically analyze an area, identify houses and streets, and suggest appropriate locations for infrastructure elements such as fiber cables, telephone poles, street cabinets, and manhole covers. The final output will be structured in a format suitable for export into Kuwaiba for network inventory management.

A key requirement of the project is ensuring that each property can be uniquely identified. In the UK, this is achieved through the Unique Property Reference Number (UPRN), which is provided by Ordnance Survey datasets such as AddressBase. While OpenStreetMap provides detailed geographic information about buildings and roads, it does not inherently include UPRNs. Therefore, part of the project involves researching how UPRN data can be retrieved from Ordnance Survey, understanding the format in which it is supplied, and determining how it can be linked to OpenStreetMap building data.

This document focuses specifically on investigating the retrieval, structure, and potential integration of UPRN data, forming a foundation for later stages of the project where automated network planning and data export will be implemented.

2. Why UPRN matters

The Unique Property Reference Number (UPRN) plays a critical role in ensuring that each property can be accurately and consistently identified within a dataset. Unlike an address, which may change over time or be written in different formats, a UPRN is a permanent and unique numerical identifier assigned to a specific location. This makes it significantly more reliable for linking datasets and managing infrastructure information.

In the context of this project, UPRNs provide a stable method of identifying individual houses when designing a telecommunications network. Since the aim is to analyze streets and residential properties in order to suggest fiber infrastructure, it is essential that each building can be uniquely referenced. Relying solely on building geometry or address text may introduce ambiguity, particularly in cases such as apartment blocks, newly built properties, or areas where address formatting varies.

UPRNs also enable integration between different data sources. For example, Ordnance Survey datasets may contain authoritative property identifiers, while OpenStreetMap provides detailed geographic building outlines. By linking these sources through spatial matching and attaching the corresponding UPRN to each building, the system gains a consistent and traceable property reference. This improves data integrity, reduces duplication, and supports future scalability of the network planning system.

Furthermore, using UPRNs aligns the project with industry and government standards, as they are widely adopted across utilities, local authorities, and telecommunications providers. Incorporating UPRNs therefore strengthens the realism and practical applicability of the proposed solution.

2. What is a UPRN

A Unique Property Reference Number (UPRN) is a permanent numerical identifier assigned to every addressable property in the United Kingdom. It is issued as part of the national address database and remains consistent throughout the lifecycle of a property, even if the address itself changes. This means that while street names, postcodes, or property names may be updated over time, the UPRN remains fixed.

Each UPRN uniquely identifies a specific geographic location, whether that is a residential house, flat, commercial building, or other addressable structure. This eliminates ambiguity that can arise from similar or duplicated addresses and ensures that each property can be referenced precisely.

UPRNs are widely used across government departments, local authorities, utility providers, and telecommunications companies. They serve as a standard reference point for managing services, infrastructure, billing, planning, and regulatory processes.

Why UPRNs Are Useful

The use of UPRNs provides several important benefits:

Prevents duplication: Since each property has one unique identifier, datasets can avoid repeated or conflicting entries for the same location.

Enables data integration: Different organizations may hold separate datasets (e.g., planning data, utility records, telecom infrastructure). The UPRN acts as a common key that allows these datasets to be linked accurately.

Improves planning accuracy: When designing infrastructure such as fiber networks, having a unique identifier for each property ensures that service coverage, connections, and resource allocation can be managed systematically and reliably.

In the context of project, incorporating UPRNs strengthens data consistency and supports accurate modelling of telecommunications infrastructure at a property level.

3. Ordnance Survey Data

Ordnance survey has several products as below.

1. Addresses and names –
Explore mapping data relating to addresses, UPRNs, named features, and places.
Discover address details, postcodes, identifiers and features.
2. Maps and imagery –
Explore mapping data designed to provide visual context and simple analysis. Discover raster maps, imagery and topography layers.
3. Transport networks –
Explore mapping data designed to provide information about roads, paths, trams, ferries and railways. Discover networks, features and identifiers.
4. Buildings and infrastructure –
Explore mapping data designed to provide information about the built environment.
Discover building and structure features, identifiers and topography layers.
5. Land and terrain –
Explore mapping data designed for land cover, land use and elevation. Discover features, greenspace and topography layers.
6. Water –
Explore mapping data designed for water bodies, tidelines and river networks. Discover features, identifiers and topography layers.
7. Areas and zones –
Explore mapping data designed to provide information about areas used for administration and analysis. Discover postcodes, boundaries and built up areas.
8. OS Net positioning data –
Discover OS Net data, providing Global Navigation Satellite Systems (GNSS, previously referred to as GPS) data.

From above available services under Addresses and names service, The UPRN can be extracted from the below listed datasets (OS products).

1. OS GB Address –

A complete and authoritative addressing dataset for Great Britain, providing a detailed view of an address and its lifecycle, based on pre-build, built, and historical property lifecycle phases.

What OS GB Address provides

UPRN data management –

Link data to an address using the Unique Property Reference Number (UPRN). This identifier is assigned to a property and persists through its full lifecycle. From construction through to its demolition, the UPRN ensures accurate and consistent management of data, even when changes are made to an address.

Data Structure – Vector

Format – CSV, GeoPackage

2. AddressBase Core –

AddressBase Core contains more than 33 million addresses. It includes Unique Property Reference Numbers (UPRNs), property-level coordinates and secondary classifications. This makes it suitable for analysis and understanding of property types, as well as uses that combines authoritative data from local authorities not found in Royal Mail Postal Address File.

What AddressBase Core provides

The dataset contains –

AddressBase Core contains more than 33 million addresses. It includes Unique Property Reference Numbers (UPRNs), property-level coordinates and secondary classifications.

Data Structure – Vector

Format – CSV, GeoPackage

3. OS Open UPRN –

OS Open UPRN provides a coordinates reference point and Unique Property Reference Number for every addressable location across Great Britain. A UPRN is an identifier used to identify an addressable location, it helps to reduce ambiguity and ensures accurate identification of locations.

What OS Open UPRN provides

Unique Property Reference Number –

A Unique Property Reference Number (UPRN) is a unique numeric identifier for every addressable location in Great Britain, found in OS's AddressBase products.

Addressable features –

UPRNs are assigned to approximately 40 million addressable locations, which may be any kind of building; residential or commercial. It may also be an object that might not have a 'postal' address – such as a bus shelter or electricity sub-station.

Share and link data –

OS Open UPRN will enable you to share and link data related to UPRNs, which you can spatially analyse and visualise using the accurate location.

Authoritative source –

The UPRNs in OS Open UPRN are primarily allocated by Local Authorities, under their legal duty to maintain this information. This means you can have confidence you're accessing an authoritative source of these identifiers.

Identifiers you can trust –

The UPRN is a persistent identifier to an addressable location that is never re-used, allowing you to confidently manage associated information to an address.

Data Structure – Vector

Format – CSV, GeoPackage

4. Methods of Retrieval

There are several ways this data can be obtained:

1. Direct Download (Licensed Access)

Licensed datasets such as OS GB Address and AddressBase Core can be accessed through formal agreements with Ordnance Survey. This may involve institutional access via a university licence or organisational subscription. These datasets are typically downloaded as bulk files in CSV or GeoPackage format.

2. OS Data Hub API

Ordnance Survey provides access to certain datasets via its Data Hub platform, which allows programmatic access through APIs. This method is suitable for automated workflows or applications that require dynamic querying rather than full dataset downloads.

3. Bulk Dataset Download

Some datasets, particularly OS Open UPRN, can be downloaded as bulk vector files. This approach is appropriate for large-scale processing where spatial matching and analysis will be performed locally.

Licensing Considerations

It is important to note that not all Ordnance Survey datasets are fully open. While OS Open UPRN is available under open data terms, products such as OS GB Address and AddressBase Core require appropriate licensing. Usage restrictions may apply regarding redistribution, commercial use, and data sharing.

For this project, licensing constraints must be considered when selecting a dataset. If the goal is to demonstrate feasibility within an academic context, OS Open UPRN may be sufficient. However, for more detailed lifecycle or classification analysis, a licensed dataset such as AddressBase Core may be required.

5. Data Conversion

Ordnance Survey datasets are typically provided in structured vector formats such as CSV and GeoPackage. While these formats are suitable for storage and spatial analysis, they may not directly align with the formats required by the proposed system architecture. Therefore, data conversion and preparation form an important step in the workflow.

Input Formats

The relevant datasets (such as OS Open UPRN, AddressBase Core, or OS GB Address) are commonly delivered in:

- **CSV (Comma-Separated Values)** – Tabular format containing UPRN, coordinates, and associated attributes.
- **GeoPackage (GPKG)** – A spatial database format that stores vector geometries along with attribute data.

Both formats contain vector-based spatial data, meaning that each record represents a geographic feature (e.g., a point location for a property).

Required Output Formats

Depending on the system design, the data may need to be converted into formats more suitable for:

- Spatial analysis (e.g., GeoJSON)
- Database storage (e.g., PostGIS)
- Integration with OpenStreetMap-derived data
- Export to downstream systems such as Kuwaiba

For example, if spatial matching is performed in a GIS workflow, converting CSV data into a spatial format (by assigning latitude and longitude as geometry) may be necessary. Similarly, GeoPackage files may need to be transformed into GeoJSON for use within web-based mapping tools.

Conversion Tools and Methods

Several established tools can support this process:

- **GDAL / ogr2ogr** – For converting between spatial formats (e.g., GeoPackage to GeoJSON).
- **QGIS** – For visual inspection, re-projection, and format conversion.
- **Python libraries (e.g., Pandas, GeoPandas)** – For programmatic processing and spatial operations.
- **Database systems (e.g., PostGIS)** – For storing and querying large datasets efficiently.

During conversion, it is important to ensure:

- Coordinate reference systems (CRS) are consistent between datasets.
- Geometry types are preserved correctly (e.g., points for UPRN locations).
- Attribute fields such as UPRN values remain intact and unmodified.

Data Preparation for Linking

Before linking Ordnance Survey data with OpenStreetMap building data, preprocessing may include:

- Removing unnecessary attributes to reduce dataset size.
- Ensuring coordinate precision is consistent.
- Filtering to a defined geographic boundary (study area).
- Standardising projection systems to avoid spatial misalignment.

Proper data conversion and preparation ensure compatibility between datasets and improve the reliability of the spatial matching process in later stages of the project.

6. Linking UPRN to OpenStreetMap

This section outlines how UPRN data obtained from Ordnance Survey can be linked to building data extracted from OpenStreetMap (OSM). Establishing this connection is essential in order to uniquely identify each property within the proposed telecom network model.

The Problem

Although both datasets describe real-world properties, they contain different types of information:

- **OpenStreetMap** provides detailed building geometries (polygons) and street layouts but does not typically include UPRNs.
- **Ordnance Survey datasets** (such as OS Open UPRN or AddressBase products) provide authoritative UPRNs and property-level coordinates, but do not include full building footprint geometries in the same way as OSM.

As a result, the two datasets cannot be directly joined using a shared attribute key. A spatial method of linking is therefore required.

Proposed Solution: Spatial Matching

The most appropriate approach is to match the datasets based on geographic location. Since both datasets include coordinate information, a spatial proximity method can be used to associate each UPRN point with the correct OSM building polygon.

Proposed Method

The linking process would follow these steps:

1. **Extract UPRN Coordinates**
Retrieve the latitude and longitude values for each UPRN record from the Ordnance Survey dataset.
2. **Load OSM Building Data**
Extract building polygons from OpenStreetMap for the same geographic area.
3. **Identify the Nearest Building Polygon**
For each UPRN coordinate point, calculate the nearest OSM building polygon using spatial analysis tools.

4. **Apply a Distance Threshold**

If the UPRN point lies within, or is within a defined threshold distance (e.g., 2–5 metres) of a building polygon, it can reasonably be assumed to correspond to that building.

5. **Attach UPRN as an Attribute**

Once matched, the UPRN can be stored as an attribute of the corresponding OSM building record in the enhanced dataset.

Considerations and Limitations

This process is known as **spatial matching** or **proximity-based spatial joining**. While it is a logical and widely used approach in GIS workflows, it is important to acknowledge certain limitations:

- Coordinate accuracy may vary between datasets.
- Some buildings may be missing or inaccurately mapped in OSM.
- Multi-dwelling buildings (e.g., apartment blocks) may contain multiple UPRNs for a single building polygon.
- Very dense urban environments may increase the risk of incorrect nearest-neighbour matches.

For the purposes of this project, this method is considered acceptable as a prototype-level solution. The objective is to demonstrate technical feasibility and logical integration between datasets rather than to achieve production-grade precision.

By implementing spatial matching, the project can successfully combine authoritative property identifiers with detailed building geometry, forming a unified dataset suitable for telecom network planning and export into systems such as Kuwaiba.

7. Challenges and risks

While the proposed approach is technically feasible, several challenges and risks must be considered when integrating Ordnance Survey UPRN data with OpenStreetMap building data. Identifying these early helps ensure that appropriate mitigation strategies can be planned in later development stages.

Coordinate Accuracy Differences

Although both datasets contain geographic coordinates, they may differ in precision and positional accuracy. Ordnance Survey data is authoritative and typically highly accurate, whereas OpenStreetMap data is community-generated and may vary depending on the area. Small positional discrepancies could result in incorrect spatial matches if tolerance thresholds are not carefully defined.

Missing or Incomplete Buildings in OSM

OpenStreetMap coverage varies geographically. In some areas, building footprints may be incomplete, outdated, or entirely missing. This could prevent certain UPRNs from being successfully matched to a corresponding building polygon, reducing overall completeness of the linked dataset.

Licensing Restrictions

Not all Ordnance Survey datasets are open. Products such as AddressBase Core and OS GB Address require appropriate licensing agreements. Usage restrictions may limit redistribution or integration with openly shared datasets. Careful consideration must be given to licensing terms to ensure compliance, particularly if the project were to move beyond an academic prototype.

Multiple UPRNs per Building

Certain properties, such as apartment blocks or subdivided commercial units, may contain multiple UPRNs within a single building footprint. Since OpenStreetMap typically represents such properties as a single polygon, additional logic may be required to manage one-to-many relationships between UPRNs and buildings.

Performance and Scalability

Handling large datasets—potentially tens of millions of address records—may introduce performance challenges. Spatial matching operations can be computationally intensive, particularly when processing large geographic areas. Efficient indexing, spatial databases, or batching strategies may be required if the system is scaled beyond a small study area.

8. Conclusion

This research demonstrates that UPRN data can be reliably retrieved from Ordnance Survey datasets such as OS Open UPRN and AddressBase products. The data is available in structured vector formats, including CSV and GeoPackage, making it suitable for processing and integration within a GIS-based workflow.

The datasets can be converted and standardised where necessary to ensure compatibility with OpenStreetMap building data. Through spatial matching techniques, UPRN coordinate points can be linked to corresponding OSM building polygons, creating a unified dataset that combines authoritative property identifiers with detailed spatial geometry.

This integration provides a strong foundation for automated telecom network planning. By uniquely identifying each property within the study area, the system can support accurate infrastructure design and enable structured export into network inventory platforms such as Kuwaiba.